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MODIS and ASTER synergy for characterizing thermal volcanic activity

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ABSTRACT

Volcanic activity is diverse in its manifestations and spans wide temporal and spatial scales. Monitoring volcanoes with any single sensor system can only provide a limited perspective on the nature of such activity because of trade-offs between spatial, temporal and spectral resolution. Spaceborne observations of volcanoes are thus optimized by utilizing data from different and complementary remote sensing instruments. This study examines the combined use of the Moderate Resolution Imaging Spectroradiometer (MODIS) and the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) for analyzing thermal anomalies from four separate volcanoes: Erta 'Ale (Ethiopia), Kīlauea (Hawai'i), Láscar (Chile) and Kliuchevskoi (Russia). MODIS provides a high temporal resolution (daily) but low spatial resolution (~1 km) dataset and is used to measure volcanic thermal radiance. ASTER provides relatively high spatial resolution (~90 m) image data and is used to define the size of thermal features, albeit at a poorer temporal frequency. Erta 'Ale and Kīlauea have been erupting continuously over the 11-year study period and their thermal emission signatures reveal a persistent baseline on which occasional increases in thermal output, associated with more profuse effusions of lava, are superimposed. Kliuchevskoi also displays occasional peaks in thermal activity due to the eruption of lava however intervening periods are characterized by weak or absent thermal anomalies. Láscar, on the other hand, provides an example of volcanic activity that is difficult for MODIS-type sensors to detect. It is characterized by fumarolic activity that is too low in temperature and too spatially limited to be readily detectable with its coarser instantaneous field of view (i.e., pixel size). However, high resolution data from ASTER can be used to supplement and complement the MODIS dataset. Beyond characterizing the size of thermal anomalies ASTER can also provide constraints on their shape, location and orientation and is capable of detecting relatively subtle thermal anomalies given its high spatial resolution thermal infrared wavebands. These qualities permitted the identification of a thermal precursor to lava flows at Kliuchevskoi. The precursor, which was associated with a rising lava lake, is characterized as an increase in the size and intensity of ASTER-detected thermal anomalies situated in the active crater. Wavelet analysis of the MODIS dataset was also carried out. This revealed the time scales over which radiant output waxed and waned during individual eruptive episodes as well as the repose time between episodes. Such information might be useful in forecasting the likely duration and onset of future eruptions.

1. Introduction

Volcanoes can be hazardous to both people and property yet also provide valuable insights into dynamic processes on Earth. It is necessary to monitor active volcanoes to mitigate the risks that they pose and to improve our understanding of these complex physical systems.

Volcanic activity involves the transfer of mass and heat to the Earth's surface. The surface temperature and emissivity of a physical body determine the amount of energy it radiates at any given wavelength. Planck's law describes this relationship, which is modified to account for spectral emissivity as follows:

$$L(\lambda, T) = \varepsilon_\lambda c_1 \lambda^{-5} / \left[\exp\left(\frac{c_2}{\lambda T}\right) - 1 \right] \tag{1}$$

where $L(\lambda,T)$ is spectral radiance (W m^{-2} sr^{-1} μm^{-1}) as a function of wavelength (λ) and kinetic temperature (T), ε_λ is spectral emissivity, and c_1 and c_2 are the radiation constants (i.e., $c_1 = 2hc^2 = 1.19 \cdot 10^{-16}$ W m^2 sr^{-1} and $c_2 = hc/k = 1.439 \cdot 10^{-2}$ m K; where h is the Planck constant, c the speed of light and k the Boltzmann constant).

This work explores the use of polar-orbiting satellites for monitoring volcanic thermal radiance from space. The types of sensors used for this purpose can be classified into two broad categories: i) high to moderate

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spatial resolution (i.e., <100 m pixel^{-1}) but low temporal resolution (i.e., >10 day) imagers, and ii) high temporal (i.e., 1≤day) but low spatial resolution (i.e., ≥1 km) imagers. The two categories will be referred to as i) high spatial resolution and ii) high temporal resolution for the remainder of this article. We note that although the meaning of adjectives such as 'advanced', 'high' and 'moderate' change with time as far as capabilities are concerned they have still been used widely to name many of the existing orbital imagers. High spatial resolution systems that have been used to monitor volcanic thermal radiance include: the Thematic Mapper onboard Landsat 4 and 5 (e.g.,, Francis & Rothery, 1987; Glaze et al., 1989; Oppenheimer, 1991; Rothery et al., 1988), the Optical Sensor on the Japanese Earth Resources Satellite-1 (JERS-1) (e.g., Dean et al., 1998; Denniss et al., 1996), the Enhanced Thematic Mapper Plus on Landsat 7 (e.g., Flynn et al., 2001; Patrick et al., 2004; Wright et al., 2001), the Advanced Land Imager on the Earth Observing-1 mission (e.g., Donegan & Flynn, 2004), and the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) onboard the Terra satellite (e.g., Murphy et al., 2011; Rose & Ramsey, 2009; Vaughan & Hook, 2006). On the other hand high temporal resolution polar orbiters include: the Advanced Very High Resolution Radiometer (AVHRR) flown on the NOAA satellite series (e.g., Harris et al., 1997; Tramutoli, 1998), and the Moderate Resolution Imaging Spectroradiometer (MODIS) (e.g., Kervyn et al., 2008; Koeppen et al., 2011; Wright et al., 2002; Wright et al., 2004) onboard both the Terra and Aqua satellites. The temporal and spectral resolution of MODIS, as well as its global coverage and reliable geometric and radiometric calibration, led to its application in the world's first automated system for global volcano monitoring, i.e., MODVOLC (Wright et al., 2002; Wright et al., 2004).

Volcanic phenomena span a wide range of temporal and spatial scales. It may take years, decades or longer for magma to migrate to the surface, whilst a pyroclastic flow can be emplaced in minutes. A volcanic ash plume can extend over thousands of kilometers yet ejecta are typically meter scale or less. This disparity means that there is no single sensor in existence that is capable of acquiring data over the spatial and temporal scales exhibited by all of Earth's volcanoes. Volcano monitoring efforts therefore need to integrate and interrogate data from networks of sensors. From a volcanologist's perspective an ideal network would include terrestrial, airborne and spaceborne sensors. In practice, since most of Earth's potentially active and active volcanoes are subject to at best only rudimentary ground-based monitoring, orbital sensors are a crucial data source for tracking global volcanism.

The Terra satellite, which carries both the ASTER and a MODIS system, offers an excellent opportunity to explore the potential of combining both high spatial and high temporal resolution datasets for monitoring volcanoes from space (e.g., Hirn et al., 2008; Vaughan & Hook, 2006; Vaughan et al., 2008). The ASTER sensor consists of three separate subsystems in the visible and near-infrared (VNIR), short-wave infrared (SWIR) and thermal infrared (TIR) with respective pixel sizes of 15, 30 and 90 m. The SWIR subsystem ceased to provide useful Earth observation data in 2008 due to a cooler malfunction although the VNIR and TIR instruments continue to operate normally. MODIS has 32 spectral channels in the VNIR, SWIR, mid infrared (MIR) and TIR wavelengths with a nominal pixel resolution of 1 km and a temporal resolution of 24 h. Multiple simultaneous viewing opportunities are possible as both sensors are carried by the same satellite platform.

The aim of this work is to compare how two different sensor classes report the temporal, spatial and thermal characteristics of a range of target volcanoes. The targets were chosen because they display different eruptive characteristics and would therefore be useful in testing the performance of high temporal resolution (i.e. MODIS class) and high spatial resolution (i.e. ASTER class) sensor systems at cataloging different styles of volcanic activity. The term "style" here is simplified to mean whether thermal activity is any of the following: low temperature, high temperature, small, large, persistent or not persistent. A decadal time series has been used to a give a longer term perspective on this.

2. Study areas

Four different volcanoes have been investigated: i) Kīlauea (Hawai'i), ii) Láscar (Chile), iii) Erta 'Ale (Ethiopia) and iv) Kliuchevskoi (Kamchatka, Russia). Each site represents a markedly different type of volcanic setting: i) intra-plate 'hot spot' volcanism with frequent effusive basaltic eruptions (Kīlauea), ii) a continental-oceanic subduction zone with persistent degassing, sporadic lava dome growth and intermittent explosions (Láscar), iii) a continental rift zone with an active lava lake (Erta 'Ale) and iv) a volcanic arc (Kliuchevskoi). These sites were selected for their contrasted settings and eruptive styles, enabling a more thorough evaluation of the capabilities and limitations of each sensor.

3. Metrics for volcanic thermal activity

This study was concerned with the characterization of thermally anomalous surfaces. Two metrics that quantify different aspects of such surfaces are defined. The first (MOD*) uses MODIS data to describe spectral intensity. The second (AST*) estimates the size of anomalous areas using the ASTER sensor.

3.1. MOD (high temporal resolution volcanic radiance)*

The MODIS dataset used in this study is comprised of all the available nighttime, calibrated radiance scenes (i.e., MOD02 1km) and their corresponding geolocation products (i.e., MOD03) acquired for each volcano by MODIS-Terra since its launch in 1999. These data are available from the Level 1 and Atmosphere Archive and Distribution System (LAADS) maintained by the NASA Goddard Space Flight Center. Over the study period (i.e., 1 January 2000 to 1 January 2012) there are 3709 scenes for Erta 'Ale, 5391 scenes for Kliuchevskoi, 4261 scenes for Kīlauea and 3772 scenes for Láscar (i.e., a total of 17,133 MOD02 1KM scenes). The combined size of this data set (i.e., 1.7 Tb) was reduced by cropping scenes and selecting pertinent spectral channels.

The MOD* metric is designed to represent the overall amount of MIR radiance from thermally anomalous areas. Given the IFOV of the MODIS instrument the vast majority of thermal anomalies will be subpixel in size. A radiance measurement from a single detector element will therefore often contain contributions from volcanically active areas and non-active background areas. We have used channel 21 (i.e., 3.929–3.989 μm) to characterize the radiance from volcanically active areas and channel 32 (i.e., 11.77–12.270 μm) to characterize the radiance from non-active background areas. This is because MIR detector arrays (e.g., channel 21) are much more sensitive to volcanic temperatures (i.e., hundreds of degrees Celsius) than TIR detector arrays (e.g., channel 32). To illustrate this point let us consider an isothermal background surface at 20 °C that occupies the entire IFOV of detector elements in channel 21 and 32; the radiance from each channel will be 0.5 and 8.1 W m^{-2} sr^{-1} μm^{-1}, respectively. In the case that the IFOV also contained a second isothermal surface with a temperature of 500 °C and area of 100 m^2 (i.e., 0.01% of MODIS pixel) then the radiance from each channel would be 11.72 and 9.32, respectively. That is, the TIR radiance in channel 32 will increase by 14.8% whilst the MIR radiance in channel 21 will increase by 2232%. The MOD* metric uses this relationship to define anomalous radiance in channel 21 as follows:

$$MOD^* = \sum_{i=1}^{n} L_i - L_{i,_{bkgd}} \tag{2}$$

where n is the number of anomalous pixels, L_i is the spectral radiance, and $L_{i,bkgd}$ the background spectral radiance, of the ith anomalous pixel in channel 21. The background radiance was calculated by using the brightness temperature in channel 32 as a proxy for the temperature of the non-active background areas. This background temperature (i.e., from TIR wavelengths) was then used to calculate background radiance (i.e., in MIR wavelengths) for channel 21. A pixel is considered to be anomalous if its Normalized Thermal Index (i.e., NTI) exceeds a threshold value. The NTI (Eq. (3)) utilizes the relative sensitivity of MIR and TIR wavelengths outlined above to detect thermally active volcanic targets from space (Wright et al., 2002).

$$\mathrm{NTI} = \frac{L_{22} - L_{32}}{L_{22} + L_{32}} \tag{3}$$

where L_{22} and L_{32} are spectral radiance measurements from MODIS channels 22 and 32, respectively. MODIS channels 22 and 21 cover the same spectral region but have different saturation limits (i.e., brightness temperatures of around 60 and 225 °C, respectively) and radiometric accuracy (i.e., noise equivalent temperature variation, NEΔT, of 0.07 and 2.0 K, respectively). If saturation occurs in channel 22 then radiance from channel 21 was automatically used instead. A global threshold of −0.8 was proposed by Wright et al. (2004) to identify volcanic thermal anomalies from nighttime MODIS scenes. This global threshold was set relatively high to avoid false alarms. To increase anomaly detection sensitivity it is possible to define volcano-specific thresholds (e.g.,Kervyn et al., 2008; Koeppen et al., 2011). To find an optimal local NTI threshold we first calculated the frequency distribution of all NTI values for each volcano (Fig. 1).

The NTI values that correspond to volcanic hot-spots represent a tiny percentage of pixels at the extreme right hand tail of the distribution, pixels that are not anomalous will correspond almost exclusively to NTI values in the rest of the distribution. An optimal threshold needs to be found to limit the number of false positives and missed anomalies. This was achieved by checking that the spatial distribution of anomalies was reasonable throughout the time series (Fig. 2) and that anomalies did not display seasonal variations through time (Fig. 3). The bimodality of the Kīlauea histogram (Fig. 1B) is due the presence of both land and water in the area used to calculate local NTI values at this volcano. The locations of anomalous pixels were recorded on a scene by scene basis and anomaly frequency maps were calculated for each volcano (Fig. 2).

These maps depict the frequency with which an anomalous NTI value was detected at a given location over an 11 year period. Zero frequency areas are filled in using MODIS visible wavebands to provide a geographic context. Local NTI histograms (Fig. 1) were calculated from the entirety of scenes depicted in Fig. 2, the MOD* time series was calculated from smaller areas outlined using a white, dashed box. Fig. 2A shows that activity at Erta 'Ale is confined to the summit caldera. The fairly large (i.e., several km^2) but low frequency region to the NNW is associated with the Alu-Dalaffilla fissure eruption of November 2008. Activity at Kīlauea is more spatially extensive and mainly corresponds with the lavas erupted on the volcano's East Rift Zone (Fig. 2B). Dotted around the Big Island are areas which contained short-lived thermal activity (i.e., low probability areas). They have previously been interpreted as resulting from vegetation fires (Koeppen et al., 2011). This might also be the cause of similar low probability areas surrounding Kliuchevskoi (Fig. 2C). Volcanic anomalies can be observed over both Kliuchevskoi and Bezymianny. Láscar's thermal features are constrained solely to the active summit crater (Fig. 2D). Clearly anomaly frequency maps display the likely location of thermally anomalous areas over time, as such, they might be useful during hazard planning exercises. They also help more generally to distinguish effusive activity (i.e., Kīlauea) from lava domes or lakes situated in craters (i.e., Láscar and Erta 'Ale). Kliuchevskoi combines both types of behavior. Thus, even if we knew nothings about these volcanoes beforehand it would be possible from Fig. 2 to determine that they exhibit different types of volcanic activity.

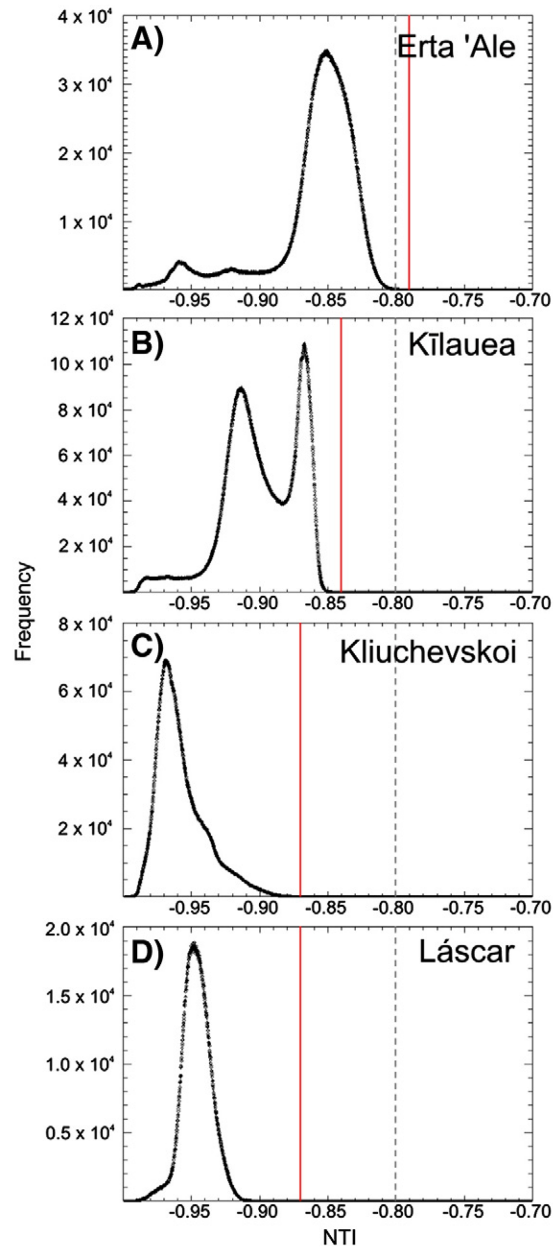

Fig. 1. The frequency distribution of MODIS Normalized Thermal Index (NTI) at A) Erta 'Ale, B) Kīlauea, C) Kliuchevskoi and D) Láscar. Optimal local anomaly thresholds are shown using a solid, red line. The global MODVOLC threshold is shown using a dashed, gray line for comparison. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Annual anomaly frequency maps for each volcano are provided in the supplementary material.

3.2. AST* (Size of anomalous areas)

This study uses the ASTER TIR at-surface radiance product (AST_09T) which is geometrically, radiometrically, and atmospherically corrected by the Land Processes Distributed Active Archive Center (LP DAAC) of the United States Geological Survey (USGS). We obtain terrestrial surface temperatures from this product using the Normalized Emissivity Method (Realmuto, 1990) with maximum emissivity set to 0.97. The ASTER Surface Kinetic Temperature (AST_08) product, which is based on the TES algorithm (Gillespie et al., 1998), was not used in this study because we could not verify whether the empirical relationship between laboratory-derived emissivity values and spectral contrast

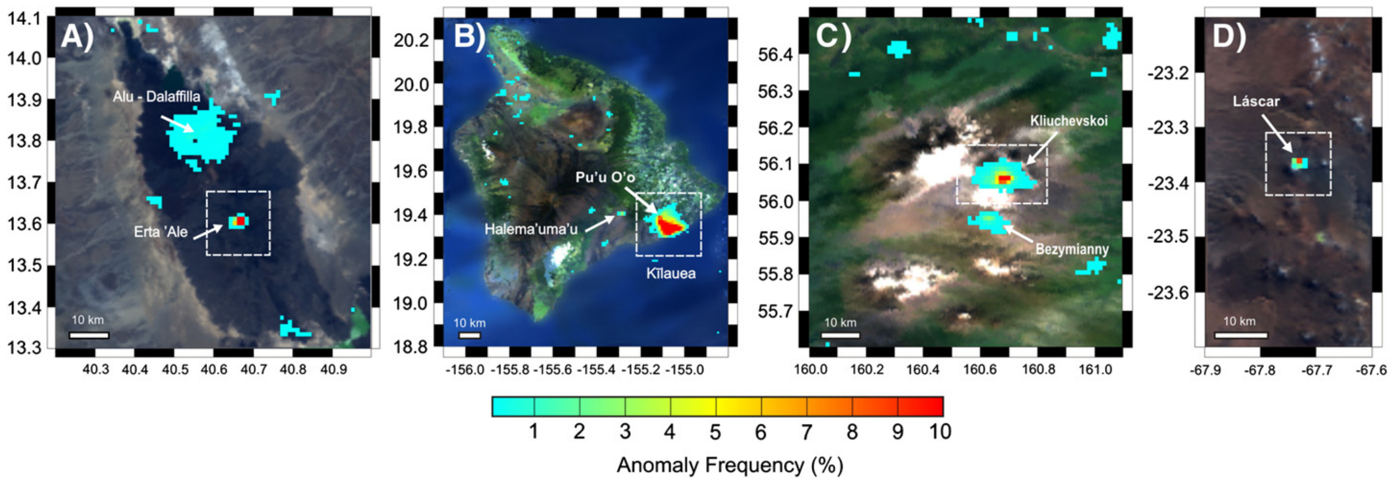

Fig. 2. Spatial distribution of anomaly frequency as detected by MODIS between 1st January 2000 to 1st January 2012 at A) Erta 'Ale, B) Kīlauea, C) Kliuchevskoi, and D) Láscar.

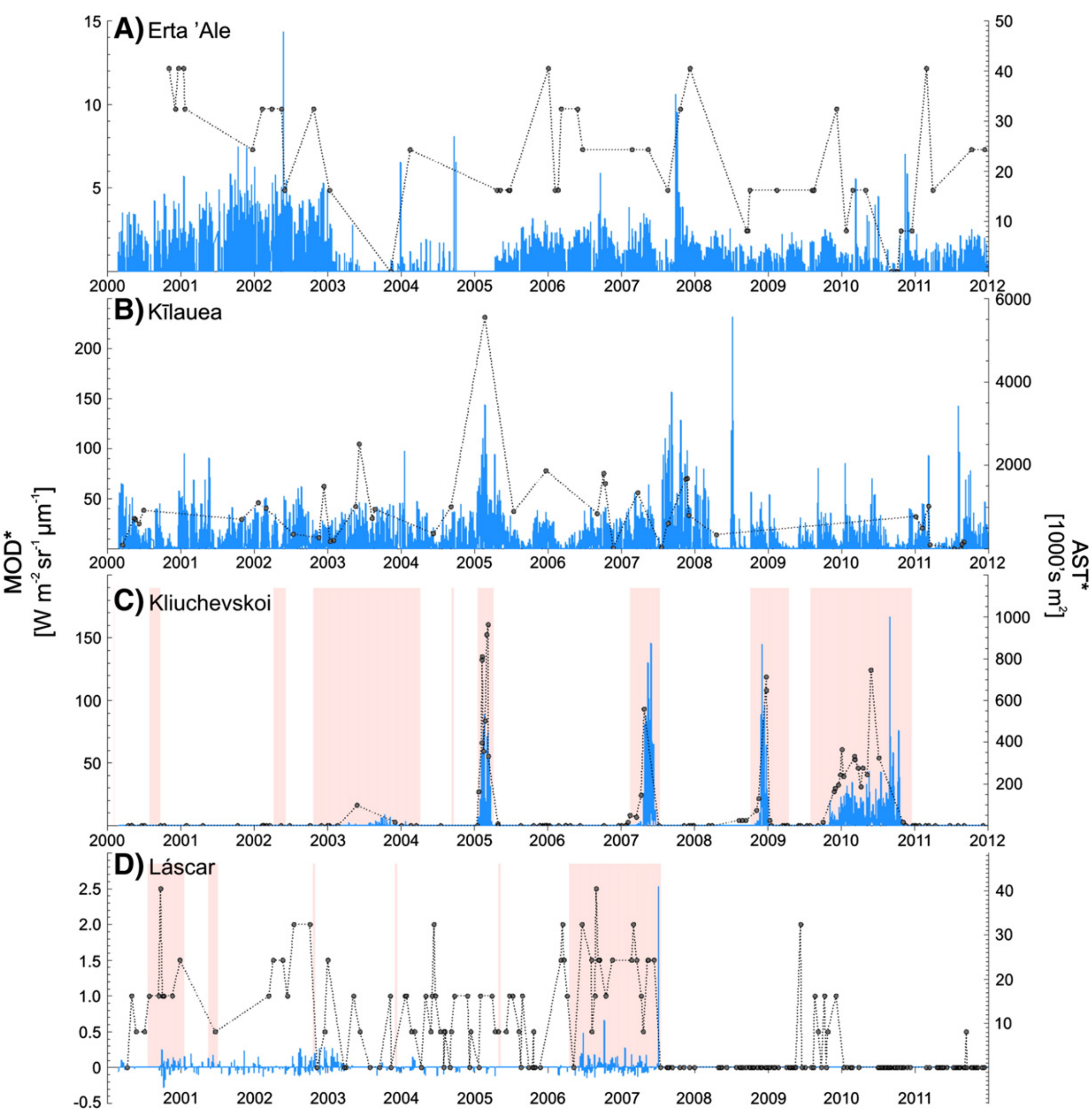

Fig. 3. The MOD* time series (blue bars) is a measure of volcanic radiance [W m^{-2} sr^{-1} μm^{-1}]; the AST* time series (filled circles with a dotted trend line) represents the size of thermally anomalous areas [1000's m^2]. Pink bars depict discrete eruptive episodes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

would apply in an active volcanic setting. An inspection for cloud cover was carried out on the temperature image. If clouds completely obscured the volcano so that it could not be located visually in a TIR image then that scene was removed from the dataset.

The AST* metric utilizes the relatively high spatial resolution of the ASTER sensor to measure the size of thermally anomalous areas.

$$AST^{*} = A.n^{*} \quad (4)$$

where A is the nominal area of an ASTER pixel (i.e., 8100 m^2) and n* is the number of anomalous pixels in a given scene. ASTER pixels were identified as anomalous if their temperature above background exceeded 40 °C. This threshold was selected to ensure an extremely low false alarm rate (i.e., perhaps zero) and to avoid the incorporation of too many ephemeral, low temperature gas plumes. The motivation for filtering cool gas plumes is that they can change dramatically in size on time scales that are orders of magnitude greater than the revisit time of the sensor (i.e., minutes compared to days or weeks) and will therefore produce scatter in the data that masks other, more persistent thermal features.

Even at the spatial resolution of ASTER TIR bands (i.e., 90 m) it is likely that its individual detector elements will measure radiance from surfaces at markedly different temperatures over hot volcanic targets. Any one pixel temperature will therefore give only a qualitative measure of volcanic thermal heating. However, given that the AST* metric is restricted to measuring the size of thermally anomalous areas, it remains robust against this effect of subpixel thermal mixing. Although anomalous pixels could be identified using pixel radiance instead of pixel temperature we have utilized the latter because it is more stable against potentially variable spectral emissivity profiles as well as wavelength dependent absorption features (e.g. due to SO_2 or ash aerosols). ASTER TIR channels will saturate quite frequently over hot volcanic targets (i.e., saturation brightness temperatures ≈ 97 °C), especially given their relatively small IFOV, this issue was also avoided by limiting the AST* metric to measuring just the size of thermal anomalies.

The temperature above background was calculated by subtracting a background temperature from the NEM temperature array. The background temperature was defined as the average pixel temperature in the background area. This area was defined as an annulus around the active crater at Erta 'Ale, Kliuchevskoi and Láscar and as a collection of user-defined points at Kīlauea. The methodology for determining the optimal size of an annulus is described in detail by Murphy et al. (2011). Hot volcanic features (e.g., lava or pyroclastic flows) occasionally crossed the background area (i.e., at Kliuchevskoi). The corresponding pixels were automatically screened from the background area using a threshold of twice the standard deviation of the entire temperature array (i.e., an area of ~62 km^2). The search area used to find anomalous ASTER pixels included (and extended beyond) the background area to ensure that all volcanic thermal features were included in the AST* metric.

Lava flows, pyroclastic flows and ash/gas plumes typically travel in a preferred direction from a stratocone due to local topography or wind direction. The 2D projection that these features create in an image acquired from space could, for illustrative purposes, be defined as a narrow arc (<20°) originating from the active crater. The relative magnitude of the area of intersection between an arc (i.e. an active volcanic feature) and an annulus (i.e. the background area) and the total area of the annulus is given by:

$$\frac{\alpha}{A} = \frac{\theta}{2\pi} \quad (5)$$

where α is the area of intersection, A is the size of the background area, and θ is the angular width of the arc in radians. For a 20° arc (i.e., 0.35 rad) the area of intersect would be 5.6% of the background area. Therefore pixels which contain contributions from cold ash plumes or warm volcanogenic material (i.e., that were not hot enough to be automatically removed by the clipping procedure) would still have a limited effect on the background temperature. The geometry of an annulus is therefore naturally stable against the intrusion of fresh volcanogenic material. Furthermore the circular nature of the annulus acts towards balancing the preferential solar heating effects that are created by varying amounts of solar irradiance around the volcanic edifice. Therefore the annulus approach for calculating background temperatures from stratocones is favored in this work. At the Kīlauea shield volcano, which frequently exhibits active lava flow fields covering several km^2, the annulus approach was not possible. The background area was therefore defined as a collection of user-defined points around the flow field in this case. We note that a single background temperature cannot accurately characterize such an extensive area given that significant, non-volcanic variations in terrestrial temperatures occur across it. The annulus methodology (i.e., used for all volcanoes but Kīlauea) is estimated to provide background temperatures to within 5 °C based on inspection of radial temperature gradients from the active crater, as decribed in Murphy et al. (2011). The user-defined points (i.e., used for Kīlauea only) give background temperatures to within 15 °C (i.e. twice the maximum standard deviation of the background data points).

3.3. Wavelet analysis

Wavelet analysis reveals the periodicity in a time series of data. A wavelet is a wave function of finite duration (i.e., it is localized). The Morlet wavelet was used with a central frequency of 6 for this study. A continuous wavelet transform will convert data from the time domain to a time-frequency domain through the convolution of a time series with translated (i.e. moved in time) and scaled (i.e. stretched) versions of the original wavelet function. This involves evaluating the integral of the product of the data points with the wavelet function. Conceptually this equates to quantifying the amount of overlap between the area under the curve defined by the data points and the area under the curve defined by the wavelet function. Power is the metric of this overlap. As power is calculated for a range of translation and scale parameters the resulting wavelet transform will represent how well the wavelet and the data match at different locations in time and at different periods of oscillation. A period of oscillation is the length of time over which an oscillation occurs (i.e., it is equivalent to the reciprocal of frequency). Power was normalized by the maximum value in each transform. This provides a unitless measure of periodicity that varies from zero to one. A practical guide to wavelet analysis is given by Torrence and Compo (1998). Examples of wavelet analysis in volcanology include studies of thermal image data from handheld infrared cameras (Spampinato et al., 2012) and geochemical time series measuring plume composition (Boichu et al., 2010; Oppenheimer et al., 2009).

4. Results

4.1. Time series

Fig. 3 shows the entire MOD*and AST* datasets. They span the period between 24 February 2000 (i.e., the first available scene) and 1 January 2012. Erta 'Ale and Kīlauea (Fig. 3A,B) have been erupting continuously over the study period. Kliuchevskoi and Láscar (Fig. 3C,D) underwent intermittent eruptive episodes that are displayed using pink shading. Eruptive histories are mostly obtained from reports of the Global Volcanism Program of the Smithsonian Institute (Venzke et al., 2002-).

Erta'Ale was frequently active throughout the 11 year study period. MOD* values are generally below 5 W m^{-2} sr^{-1} μm^{-1} (i.e., 99.3% of data points), although peaks in thermal output do occasionally exceed this value (i.e., up to a maximum MOD* of 14.4). There are two periods were MOD* values remain close to zero for months at a time (i.e., June–November 2003 and October 2004 to April 2005). This may indicate that the height of the lava lake was particularly low during these periods. Conversely, episodes characterized by high lava lake levels – as observed in the field (e.g., November 2010 overflow) – correspond with peaks in MOD* values (i.e., 7.0 in this case). Variations in lava lake height are controlled by a number of factors, including magma reservoir pressure, vesiculation and conduit–crater morphology.

The size of the thermal anomaly, as given by the AST* metric, varies between roughly 1 and 4×10^5 m^2 (i.e., 1–5 ASTER pixels). The linear Pearson correlation coefficient between AST* and MOD* is 0.54. That is, the size of thermally anomalous areas as measured by ASTER does not show a particularly strong correlation with the MIR radiance measured by MODIS. This is because the radiance emitted from the lava lake varies considerably more than the size of the lava lake. Radiant output can change significantly for a given lake area, for example, through convective overturning of the lake surface and bubble burst, both of which result in varying exposure of hot, incandescent material through time.

Kīlauea is the most consistently active volcano in this study and is characterized by frequent fissure, flank and crater eruptions of low viscosity basalt lava. These effusive eruptions often result in tube fed flows that are capable of traveling several kilometers from the source vent. Consequently the MOD* time series displays a relatively high average of 7.5 W m^{-2} sr^{-1} μm^{-1} with peaks exceeding 50 W m^{-2} sr^{-1} μm^{-1} (i.e. coincident with outbreaks from the lava tubes). Several episodes of elevated thermal emission are apparent (e.g., January–September 2005, July 2007 to April 2008, and July 2008). The latest of these is particularly significant because it was the largest in the data set (i.e., MOD* of 232; 7th July 2008), it occurred during the eruption of three simultaneously lava fountains (Smithsonian Institution, 2010) and yet was completely missed by the ASTER sensor. Although high spatial resolution imagers, such as ASTER, are more likely to miss such short lived dynamic events, the imagery that they provide offers superior constraints on the size, shape, orientation and location of thermal anomalies. For example, a lava flow field imaged simultaneously by both sensors on 7 June 2003 (Fig. 4) exhibits individual lava tubes of varying age that clearly form an interconnected network in the ASTER image (Fig. 4A), whilst it is only possible to determine the orientation and length of the overall flow field in the MODIS image (Fig. 4B). High spatial resolution data, as exemplified here by ASTER, would clearly be more useful for operational assessment of i) damage to local infrastructure (e.g., buildings, roads, etc.), ii) future flow direction (especially given that surface breakouts occur from the tubes), and iii) the risk of fire (e.g., through proximity to combustible material), given timely availability of the image, or a high-level monitoring product derived from the image. The size of thermal anomalies at Kīlauea has a mean AST* value of 8.9×10^5 m^2 (i.e., 110 pixels) with a standard deviation of 9.6×10^5 m^2 (i.e., 118 pixels). They can therefore be very large yet also vary significantly in size through time. The most extensive anomaly in the AST* series is 5.55×10^6 m^2 (i.e., 685 pixels). The correlation coefficient between AST* and MOD* is 0.65 (i.e. anomaly size and MIR radiance are more closely matched at this volcano than at Erta 'Ale).

Kliuchevskoi exhibits a more sporadic eruptive style, often spending prolonged periods of time with no, or very limited, signs of thermal activity (Fig. 3C). There are four major peaks in the MOD* series i) 20 January to 7 April 2005, ii) 22 April to 25 June 2007, iii) 30 October 2008 to 10 January 2009, and iv) 1 November 2009 to 1 November 2010. These correspond with Strombolian eruptions and lava flows (Venzke et al., 2002-). The first three major peaks are fairly symmetric and last around 2–3 months. The forth is asymmetric and last for about a year, with a prolonged period of Strombolian activity, followed by more extensive lava flows and culminating in a Vulcanian eruption (Venzke et al., 2002-) after which time thermal activity came to an abrupt end.

The maximum MOD* values for the symmetric peaks are, in consecutive order: 88.6, 145.8 and 144.9. These peaks have mean MOD* values of 10.4, 30.0 and 11.8, respectively. The forth major peak displays a higher maximum MOD* value (i.e., 166.7) and a lower mean MOD* value (6.9). The maximum extents of thermal anomalies during these four major episodes range from 5.6×10^5 to 9.6×10^5 m^2. That is, they are comparable in size to Kīlauea flows, although overall slightly smaller. Correlation between AST* and MOD* is relatively high overall (i.e. 0.85). The increased correlation between size and intensity at Kliuchevskoi compared to Kīlauea is probably because lava flows from Kliuchevskoi's stratocone do not form tubes. At Kīlauea however the formation of tubes means that radiance from these flows will depend on the proportion of exposed to concealed lava.

Inspection of ASTER scenes reveals a thermal precursor to two of the lava flow events at Kliuchevskoi (Fig. 5). The precursor is defined

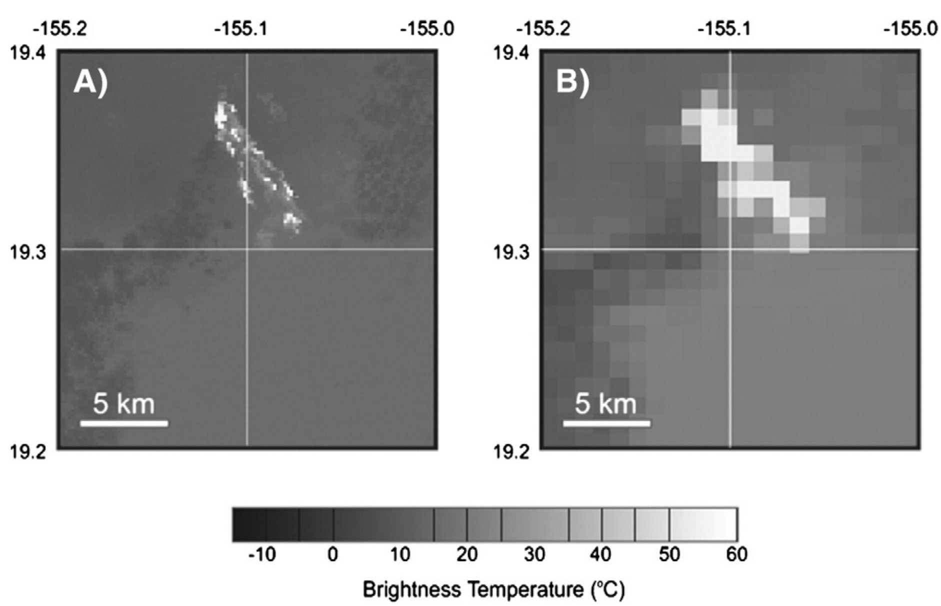

Fig. 4. Brightness temperatures from A) ASTER channel 10 and B) MODIS channel 22 of the Pu'u 'Ō'ō flow field on the 7th June 2003. ASTER is able to characterize the size, shape, location and orientation of individual lava flow tubes.

by thermal anomalies that are situated within the summit crater and that increase in both size and intensity through time. The increase in size is interpreted as resulting from an increasing amount of ponded lava in the active crater, as has been observed in the field (Smithsonian Institution, 2009). The increase in intensity is due to greater radiant heat loss from the lava surface, which is dependent on both size and temperature. After the precursory infilling of the summit crater, lava breached the crater rim and descended the flanks of the volcano (Smithsonian Institution, 2007, 2009). Such observations could be used to create simple, automated thermal precursors to lava flow activity at Kliuchevskoi and other similar volcanoes. If, for example, an alert had been based on detection of 5 or more pixels situated in the active crater with a temperature above background of at least 40 °C, then both lava flow events could have been anticipated two weeks before they were seen descending the edifice. This demonstrates the utility that moderate to high spatial resolution datasets can provide when their revisit periods are adequate to capture changing levels of activity.

Láscar displays very low MOD*, i.e., mainly lower than 0.5 and with a maximum of 2.5 (Fig. 3D). Scatter in MODIS anomalies, created by imperfect correction for MIR background radiance and atmospheric effects, can result in negative anomaly values. Thermal activity at this volcano, characterized by persistent degassing from fumaroles in and around the active crater, is particularly difficult for low spatial resolution sensors such as MODIS to detect because the thermally active region is low in temperature and relatively small in size compared to the instantaneous field of view (IFOV) of the sensor. Consequently the radiation falling on any one detector element contains a limited contribution from the target area. ASTER, on the other hand, regularly detects thermal anomalies at Láscar. AST* values range from roughly 10,000 to 40,000 m^2 in size. These anomalies are probably due to varying degrees of fumarolic activity from the active crater (Murphy et al., 2011). The higher spatial resolution (i.e. smaller IFOV) of ASTER allows it to detect relatively subtle thermal anomalies. Fig. 6 illustrates this point by displaying an ASTER image of a persistent thermal anomaly in Láscar's active crater (Fig. 6A) juxtaposed with a contemporaneous MODIS image acquired over the same area (Fig. 6B). The anomaly is evident in the ASTER image yet remains completely undetected by the MODIS sensor.

4.2. Wavelet power spectra

The high temporal resolution of MODIS lends itself to more detailed time series analysis. A continuous wavelet transform was applied to the MOD* time series of a) Erta 'Ale, b) Kīlauea and c) Kliuchevskoi (Fig. 7). Shown for each volcano are i) the original MOD* time series, and ii) the wavelet power spectrum. The axes of the wavelet power spectrum are time against period of oscillation. Time is plotted in years (x-axis) and period of oscillation is plotted in years (left y-axis) using a log base 2 scale and also in the equivalent amount of days (right y-axis). Power is represented using a color scale. Peaks in the power spectrum (i.e. warmer colors) indicate periodicity in the time series at the corresponding time and period of oscillation. During the convolution process wavelets may be translated towards (or beyond) the edge of the finite time series, especially at longer periods, producing spurious results in the corresponding part of the power spectrum. This region is known as the cone of influence and is represented in gray to leave only the reliable part of the power spectrum visible. Certain instances in time are highlighted in the power spectrum using a vertical, dashed white line. The corresponding period against power plot for each of these time slices is displayed in Fig. 8.

Erta 'Ale (Fig. 7Aii) displays two time intervals revealing a periodicity of approximately 1/2 a year centered around June 2002 and August 2007. The latter event also shows significant periodicity at 1/16 to 1/4 of a year (i.e., 23–91 days). A time slice of the power spectrum at these two times (i.e. time slice 1 and 2) is displayed in Fig. 8. These time slices also clearly show low period signals (i.e. around 2 days) that are approximately contemporaneous with sharp peaks in the MOD* time series (Fig. 7Ai). A strong periodicity (i.e., power = 0.93) occurs at a long period (i.e., 4 years) around the turn of 2006 (Fig 7Aii). It is associated with the spacing between the two periodicity events highlighted in time slice 1 and 2. This spacing is actually closer to 5 years than 4 but is not reliably definable within this power spectrum because the corresponding periodicity falls into the cone of influence. To characterize longer period events, such as this one, it is necessary to have a longer time series. Less prominent peaks also appear in the power spectrum of Erta 'Ale. Most remarkable among these are perhaps the dual periodicity component at 6.16×10^{-3} and 2.3×10^{-2} years (i.e., 2 and 8 days) because they reoccur throughout much of the time series and therefore appear to be characteristic periods of oscillation at this volcano.

Kīlauea displays strong periodicity at a period of 2.5 years with maximum power of this feature occurring around October 2006 (Fig. 7Bii). This long period signal is overlain at different times with shorter period oscillations. Time slices 3 and 4 (Fig. 8) display some of the multiple periodicity components in the Kīlauea power spectrum. The long (i.e. 2.5 year) period is the dominant feature in these time slices however both also show periodicity at around 1/4 of a year and at 2 days. Time slice 3 also shows a peak in periodicity at around 1 year which is equivalent to the duration of the overall waxing and waning in radiance in the corresponding time series (Fig. 7Bi). A strong low period oscillation (i.e. 2.6 days) in the power spectrum occurs during a short sharp peak in MOD* values around July 2008.

The four major eruptive episodes at Kliuchevskoi dominate both the time series and the wavelet power spectrum (Fig. 7Cii). The three symmetric events each show a waxing and waning of measured

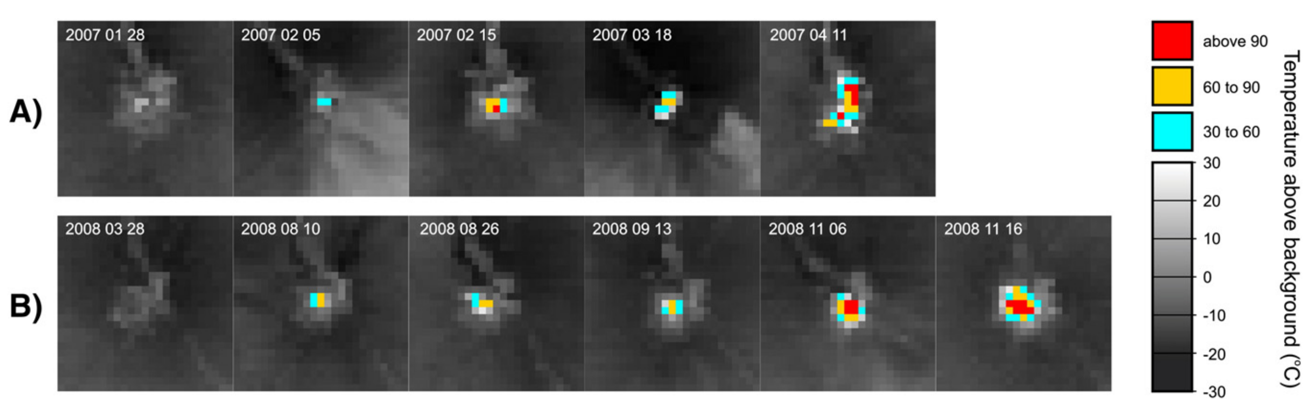

Fig. 5. Thermal precursors to lava flow activity at Kliuchevskoi are characterized by anomalies in the active crater that increase in size and intensity.

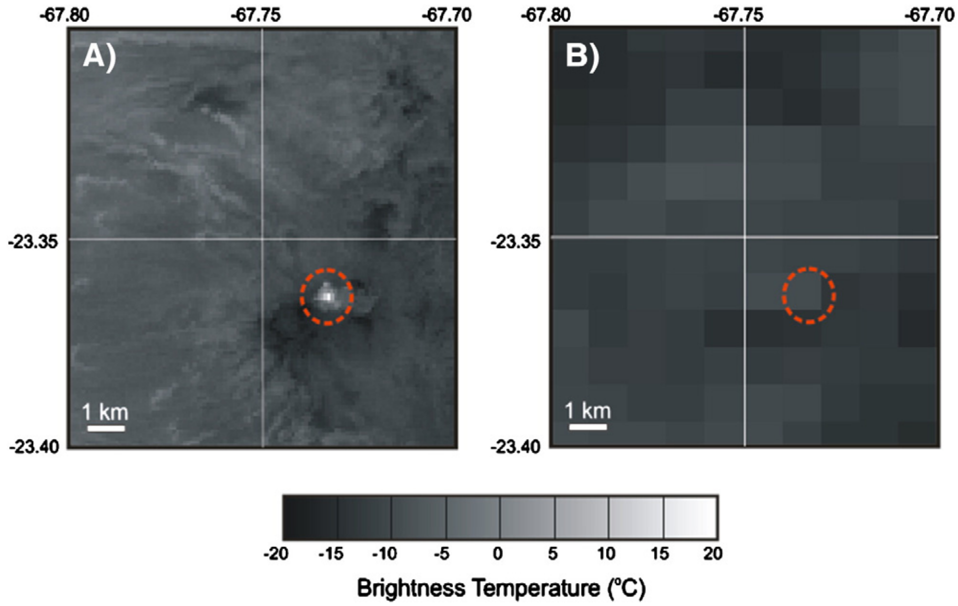

Fig. 6. Contemporaneous images of A) ASTER channel 10 and B) MODIS channel 22 of Láscar taken on 2nd October 2000. ASTER is capable of detecting subtle thermal anomalies that may fall below the MODIS detection limit. Láscar's active crater is highlighted using a red dashed circle.

radiance with a periodicity of approximately 0.25 to 0.5 years (e.g., time slice 5 in Fig. 8). The maximum periodicity event (i.e. power = 1) for Kliuchevskoi is displayed in time slice 5 (i.e., period = 3.5 days; centered on the 31st May 2007). It occurred at the same time that a new lava flow was reported to have moved down the East flank of the volcano (Smithsonian Institution, 2007). Periodicity during the episode with more persistent Strombolian activity (i.e., time slice 6) displays more clearly separated peaks in period at 2.5, 16.5, 47 and 204 days. The fairly regular spacing (i.e., repose) of the last 3 eruptive episodes resulted in a 1.5 year periodicity centered around the turn of 2009.

5. Discussion

The MODIS and ASTER sensors provide datasets with their own particular advantages and disadvantages. MODIS channel 21 will very rarely saturate, even over hot volcanic targets, and can therefore be reliably used to provide a measure of volcanic thermal radiation if the target is hot and/or large enough. The daily revisit time of the MODIS sensor also allows dynamic volcanic events to be characterized in relatively high temporal detail. This sampling frequency could be improved further still, if necessary, by inclusion of data from the other MODIS sensor (i.e., on the Aqua satellite) as might be of critical importance during a volcanic crisis. The higher spatial resolution of the ASTER sensor, on the other hand, allows the spatial characteristics of thermal anomalies to be quantified more accurately and also facilitates the detection of subtle thermal anomalies. These capabilities permitted the identification of a thermal precursor to lava flows at Kliuchevskoi. Unfortunately the probability of ASTER capturing such precursor events is limited by the fact that it is not a mapping mission (i.e. does not acquire data continuously).

The low spatial resolution of MODIS leaves the sensor incapable of monitoring the type of thermal activity expressed by Láscar during the study period (i.e., predominantly fumarolic). There are plenty of other (potentially hazardous) volcanoes around the world that are similarly difficult for MODIS to monitor. Fortunately ASTER can provide a complimentary dataset that, when used in conjunction with MODIS datasets, results in a synergy with better overall temporal, spatial and radiometric resolution. This can be used to obtain more information on a greater range of volcanic activity than either sensor could offer individually.

This is the first time to our knowledge that wavelet analysis has been applied over such a long (i.e., 11 year) data set of volcanic radiance. This enabled us to identify oscillations on weekly, monthly and yearly time scales. Very short period events (i.e., a few days) can be created by a burst of thermal activity such as from a short lived lava flow or lava fountain. It is possible, however, that similar oscillations in radiance – as seen from space – might also occur due to changes in cloud cover over a thermally stable target. Very short period events should therefore be interpreted with care and scenes should be inspected for cloud cover.

Long period events could be better resolved if the overall time series were more extensive. This reflects the need for continuous collections of data over multiple years and, eventually, decades. Fortunately the next generation of high temporal resolution polar-orbiting satellites are coming through (i.e., Suomi National Polar-orbiting Partnership (Suomi NPP), Sentinel 3, and the Global Change Observation Mission climate change series (GCOM-C). Wavelet analysis provides new constraints on the probability of the duration and repose of future events, and could be used as input into decision making tools such as the Bayesian event tree model for eruption forecasting, i.e., BET EF (Marzocchi et al., 2008).

6. Conclusion

This study investigated 11 year time series of MODIS and ASTER thermal anomalies (i.e., using the metrics MOD* and AST*, respectively). MOD* measures the radiance emitted by a volcano and AST* measures the size of thermally anomalous areas. This is the first time (to our knowledge) that two sensors of this type have been used to compare such long time series of the intensity and size of thermal anomalies. We show practical examples of the relative strengths of each sensor and how they can be used to improve overall understanding of volcanic activity at a number of distinctly different volcanoes.

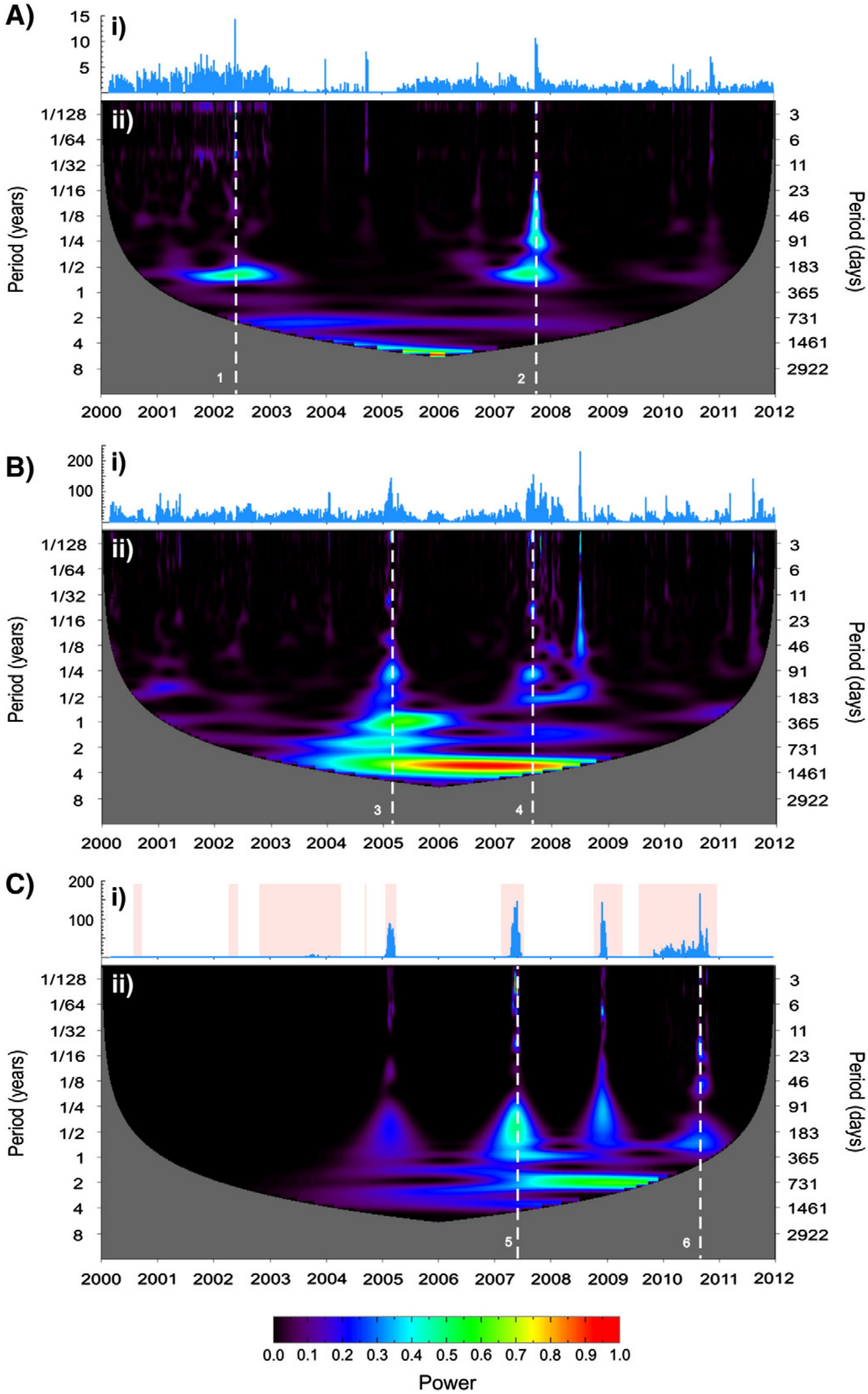

Fig. 7. Wavelet transforms of MOD* time series for A) Erta'Ale, B) Kīlauea and C) Kliuchevskoi. Shown are i) the original MOD* time series and ii) its wavelet power spectrum. Time slices of the power spectrum (shown in Fig. 8) are depicted here using dashed white lines.

The high temporal resolution MOD* metric was also analyzed using continuous wavelet transforms. Given the length of the time series it was possible to determine periods over which thermal activity waxes and wanes at a particular volcano as well as to characterize repose time between thermal events. This might be useful in guiding forecasts on the duration and onset of future eruptions. It also

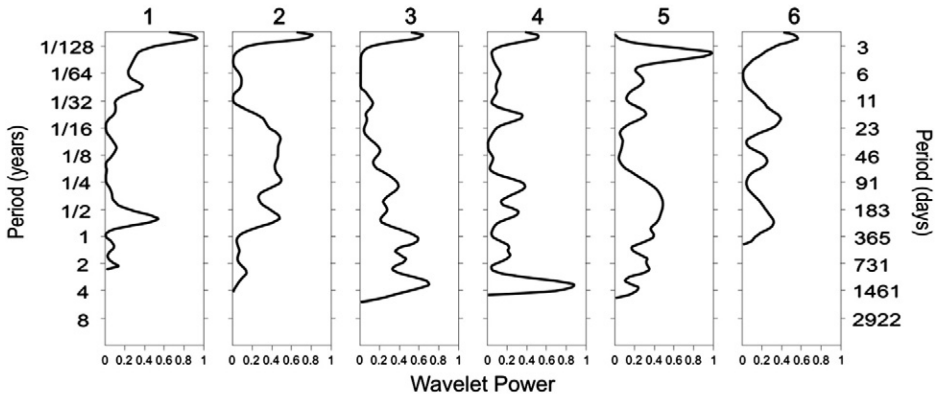

Fig. 8. Time slices of wavelet power against period.

facilitates the comparison of the time scales over which different volcanoes erupt. The AST* metric allowed the size of thermal anomalies to be determined in relatively high detail, it also facilitates the detection of subtle thermal anomalies, both of these characteristics led to the identification of a thermal precursor to lava flow at Kliuchevskoi. It is useful, therefore, to explore synergies between high temporal and high spatial resolution datasets.

Time series plots, images and movies can be thought of as volcano monitoring products. These products could be made automatically and provided free of charge to volcano observatories and other interested parties. This would increase the science return from existing and future missions, raise the global profile of the space agency or corporation that provides the data and also contribute positively to society, especially to those at risk from volcanic hazards, be it either directly (e.g., local inhabitants) or indirectly (e.g., airplane passengers and international businesses).

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Appendix A. Supplementary data

Supplementary data associated with this article can be found in the online version, at doi:10.1016/j.rse.2012.12.005. These data include Google maps of the most important areas described in this article.

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